

Nd:YAG Laser Posterior Capsulotomy

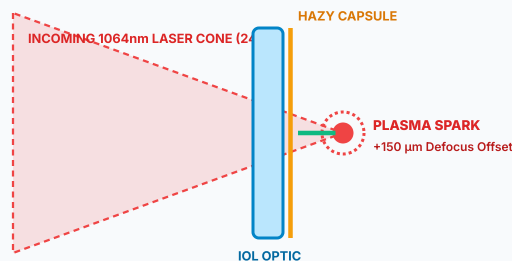
Slit-Lamp Laser Photodisruption for Posterior Capsule Opacification (PCO)

LASER WAVELENGTH	PULSE ENERGY	DEFOCUS OFFSET	CONTACT LENS
1064 nm Q-Switched Nd:YAG	1.0 – 1.8 mJ per pulse	+150 to +250 μm (Posterior)	Abraham +66D Capsulotomy Lens

☀ What Is a YAG Laser Capsulotomy in Simple Terms?

Months or years after successful cataract surgery, up to 30% of patients notice their vision getting blurry again. This is NOT the cataract coming back! Rather, microscopic leftover cells grow across the clear back bag like frost on a window pane (called an "after-cataract" or PCO). In a quick, painless 3-minute clinic procedure, we use an invisible infrared laser to zap a clear window through the frost, restoring crisp, sharp vision instantly.

LASER PHOTODISRUPTION PHYSICS & POSTERIOR DEFOCUS OFFSET (+150MM)



CRITICAL SAFETY RULE:

Why NEVER fire at 0 offset?

1. Plasma expands FORWARD back toward the incoming laser.
2. At 0 offset, the supersonic shockwave pits & cracks the IOL!

✓ +150 μm offset protects the optic 100% of the time!

STEP
1

Abraham Capsulotomy Lens Placement — Magnifying & Stabilizing the Eye

Tool: Abraham +66D Contact Lens + Gel

👉 WHAT TO DO RIGHT NOW:

Apply a drop of clear methylcellulose coupling gel to the concave face of the Abraham contact lens. Rest the lens smoothly on the patient's numbed cornea.

💡 **Why This Step Is Strictly Necessary:** The contact lens holds the patient's eyelids open to prevent blinking. Its +66 diopter central button widens the laser beam cone from 16° to 24°. This concentrates maximum laser power onto the cloudy membrane while dispersing the energy into a harmless blur as it travels to the cornea in front and the retina in the back.

⚠ **Critical Hazard:** Air bubbles trapped in the coupling gel create optical scattering and distort the laser focal point.

STEP
2

Dual HeNe Aiming Beam Convergence — Confocal Laser Alignment

Tool: Twin Red HeNe Aiming Beams (632.8nm)

👉 WHAT TO DO RIGHT NOW:

Look through the slit lamp eyepieces. Move the joystick forward or backward until the two red laser aiming dots overlap and merge into ONE single sharp, bright red dot on the cloudy membrane.

💡 **Why This Step Is Strictly Necessary:** The 1064nm laser beam is infrared—completely invisible to human eyes! To let you aim, two visible red laser beams are projected from different angles. When the two red spots merge into one, the invisible laser is focused with microscopic precision on the target tissue.

⚠️ **Critical Hazard:** Firing when the twin aiming beams are doubled causes the laser spark to detonate in the wrong plane, striking the artificial lens or anterior vitreous.

STEP
3

Posterior Defocus Offset (+150 μ m) — The Crucial Safety Gap

Setting: Defocus Knob Set to +150 μ m Posterior

👉 WHAT TO DO RIGHT NOW:

Check the laser console setting. Verify the focal offset is set to +150 to +200 micrometers posterior. NEVER fire at zero offset!

💡 **Why This Step Is Strictly Necessary:** When the laser sparks, it creates a superheated plasma bubble that expands FORWARD toward the incoming beam. If your focus is set right on the membrane (zero offset), the shockwave blasts directly into the back surface of the acrylic lens, leaving permanent pits and cracks that ruin the patient's vision with permanent glare. Setting the focus 150 microns behind the membrane allows the shockwave to cut the capsule without touching the lens.

⚠️ **Critical Hazard:** Zero offset causes irreversible pitting of the intraocular lens optic, requiring expensive surgical lens exchange.

STEP
4

Cruciate Pattern Laser Breakdown — Cutting in a Cross (+)

Tool: Q-Switched 1064nm Nd:YAG Laser (1.2–1.5 mJ)

👉 WHAT TO DO RIGHT NOW:

Fire laser pulses in a cross shape (+): start at 12 o'clock (top), then 6 o'clock (bottom), then 9 o'clock (left), and 3 o'clock (right). Cut from the outer edges toward the center.

💡 **Why This Step Is Strictly Necessary:** The cloudy membrane is under tight drum-head tension. Cutting the outer edges in a cross releases the tension, causing the four triangular flaps to curl up and roll back out of the line of sight automatically like opening stage curtains! Cutting edges first uses far less total energy and keeps the center of sight 100% safe from accidental lens strikes.

⚠️ **Critical Hazard:** Firing directly in the center first risks pitting the central visual axis. Using excessive total energy (>60 mJ) causes acute glaucoma spikes and retinal swelling (macular edema).

STEP
5

Visual Axis Clearance & IOP Check — Final Inspection & Pressure Guard

Tool: Slit-Lamp Retroillumination + Tonometry

👉 WHAT TO DO RIGHT NOW:

Inspect under retroillumination: confirm a clean, round 3.5 to 4.0mm central opening free of floating tags. Check that the gel bag behind the eye is intact, and instill one drop of apraclonidine or brimonidine.

💡 **Why This Step Is Strictly Necessary:** Loose tags of capsule hanging in the pupil will swing like pendulums when the patient moves their head, causing annoying flashes, shadows, and double vision. Furthermore, microscopic debris released by the laser can clog eye drains for several hours, causing transient pressure spikes that are safely prevented with pressure-lowering drops.

⚠️ **Critical Hazard:** Rupturing the anterior vitreous face allows vitreous gel to leak forward, increasing the risk of retinal tears and retinal detachment.